An Adaptive Multimodal AR Storytelling Framework for Primary STEM Education through Embodied Gesture and Voice Interaction

R26-IT-009

An Adaptive Multimodal AR Storytelling Framework for Primary STEM Education through Embodied Gesture and Voice Interaction

Research Question - How does using Augmented Reality Storytelling with interactive gestures, voice commands, and gamified quizzes influence and impact on the STEM academic engagement of primary school students?

Users and Stakeholders

Users

- Students
- Teachers and Educators
- Parents and Guardians

Stakeholders

- Developers / Researchers
- Industry / EdTech Sector
- Education Authorities
- Schools

Impact of the problem to user and society

Students struggle to understand abstract STEM concepts because they cannot see or interact with them in real life. Many existing digital learning tools only allow passive viewing, which can lead to boredom and low knowledge retention. As a result, students may lose interest in science at an early stage. This can impact society by reducing the number of students pursuing STEM fields in the future and widening the gap between those who have access to engaging learning tools and those who do not.

Existing Solutions and limitaions

Existing Solution	What It Does	Main Limitation
Merge EDU (Merge Cube)	Let students hold and rotate 3D models (e.g., solar system, organs)	Requires physical cube rotation or tapping; no hands-free gestures to act out concepts
CoSpaces Edu	Students code 3D stories and virtual worlds using blocks	Needs prior coding knowledge; can distract from actual STEM learning
Wonderscope	Children read and interact with stories using speech	Mostly verbal; lacks physical gesture interaction to reinforce concepts
JigSpace	Shows step-by-step 3D views of mechanical objects	Limited topics; lacks narrative depth and interactive problem-solving
Froggipedia	Virtual frog dissection and metamorphosis	Basic quizzes only; no adaptive learning or feedback loop
TimePod Adventures	Narrative episodes (e.g., Mars missions)	Short, researcher-led prototypes; limited multimodal or gesture-based interaction

Proposed solution

Our proposed solution is a multimodal AR storytelling system that helps primary school students learn complex STEM concepts through a unified "Action-Talk-Test-Play" cycle

Our solution allows students to interact with STEM concepts through hand gestures embedded in a story. Unlike existing AR tools, which mostly let students look at models or answer quizzes, our system turns learning into an active, hands-on experience. It also provides feedback and narrative progression, keeping students engaged and helping them understand the concepts deeply

AR STEM Story desgin and Embodied Gesture Interaction

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AR STEM Story desgin and Embodied Gesture Interaction

Problem Definition

Traditional AR learning tools trap students in passive observation — children watch 3D animations rather than interact with them, creating an "interaction barrier" that disconnects abstract STEM concepts from physical understanding. This project addresses that gap through embodied cognition, requiring students to physically act out STEM concepts using hand gestures to progress the story — transforming them from passive spectators into active participants.

AR STEM Story desgin and Embodied Gesture Interaction

Research gap

1. **Passive observation — Children watch 3D animations rather than participate. This system moves them from passive observers to active protagonists through physical gesture interaction.**
2. **The meaning gap — Students in self-directed learning struggle to connect visual content to actual science. Concept-mimicking gestures are required to unlock story progression, forcing active engagement with the concept.**
3. **Screen-tap dependency — Most mobile AR relies on tapping or physical prop manipulation (e.g. Merge Cube). MediaPipe skeletal tracking enables natural, hands-free scientific mimicry instead.**

App / System	Hands-Free Interaction	Skeletal Hand Tracking	Concept-Mimicking Gestures	Late Fusion (RGB + Skeleton)	Dynamic Movement Recognition
Our Storytelling app	✓	✓ 21-point MediaPipe	✓	✓	✓
Merge EDU	× (marker-based)	× (device camera only)	× (tap/tilt only)	×	Limited (tilt gestures)
Wonderscope	× (screen-tap)	×	×	×	×
CoSpaces Edu	× (controller/tap)	×	×	×	×
JigSpace	× (screen-tap)	×	×	×	×
Froggipedia	× (screen-tap)	×	×	×	×
TimePod Adventures	× (tap/drag)	×	×	×	×

AR STEM Story desgin and Embodied Gesture Interaction

Contribution to the Overall Solution

- Hands-free — students interact using only natural hand movements, no props or tapping required
- Learn by doing — students must physically act out the science concept to unlock the next story chapter
- No more passive watching — the story stays locked until the correct gesture is demonstrated

Novelty of the Component

- Gestures as Learning, Not Control — Movements directly represent scientific concepts, making interaction itself the learning experience.
- Active Story Eng by Understanding — Progression requires correct physical demonstration, turning gestures into proof of learning

AR STEM Story desgin and Embodied Gesture Interaction

Research Methodology

This component integrates knowledge from multiple domains:

1.Human-Computer Interaction (HCI)

- Designing intuitive, gesture-based interaction for natural learning.

2.Computer Vision

- Enabling real-time hand tracking and gesture recognition.

3.Augmented Reality (AR)

- Creating immersive, interactive STEM visualizations.

4.Educational Technology

- Improving engagement through embodied, concept-driven learning.

AR STEM Story desgin and Embodied Gesture Interaction

Research Objective

The main goal of this research is to create and test a hand gesture-based interaction system within an adaptive AR storytelling platform.

Sub-Objectives

- Design a STEM storytelling system where children use gestures to interact and progress through the story.
- Develop a real-time hand gesture recognition system in Unity with high accuracy in different environments.
- Integrate gesture controls with a voice-based query system to create an interactive learning flow.
- Evaluate student engagement and learning improvement through user studies and performance

AR STEM Story desgin and Embodied Gesture Interaction

Technologies Used

The system uses modern technologies including:

- **Google Mediapipe**
- **Late fusion**
- **Unity AR / ARCore / ARKit** for Mixed Reality interaction
- **Firebase**

AR STEM Story desgin and Embodied Gesture Interaction

Validation and Evaluation

The system will be evaluated through:

- User Testing (Primary Students)
- Measuring engagement and concept understanding
- Gesture Recognition Accuracy
- Evaluating precision of hand tracking and movement detection
- Learning Effectiveness
- Assessing improvement in comprehension
- Feedback Collection
- Gathering insights from teachers and students

Data Availability, Security and Ethics

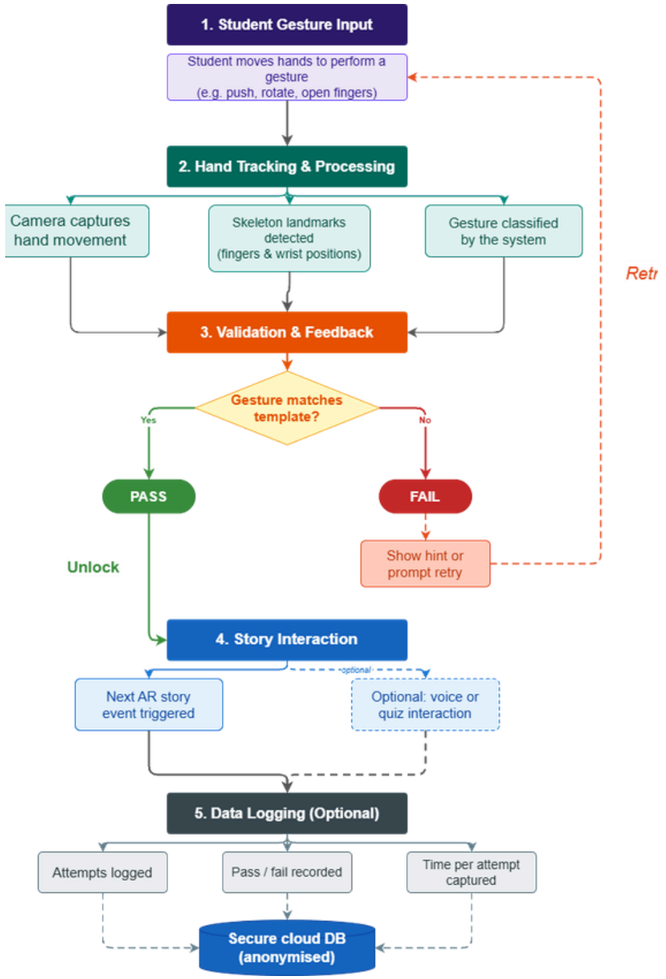
- Secure Data Processing
- Student inputs are handled safely and responsibly
- Privacy Protection
- No personal data stored without consent
- Ethical Compliance
- Follows guidelines for research involving children

AR STEM Story desgin and Embodied Gesture Interaction

System Overview

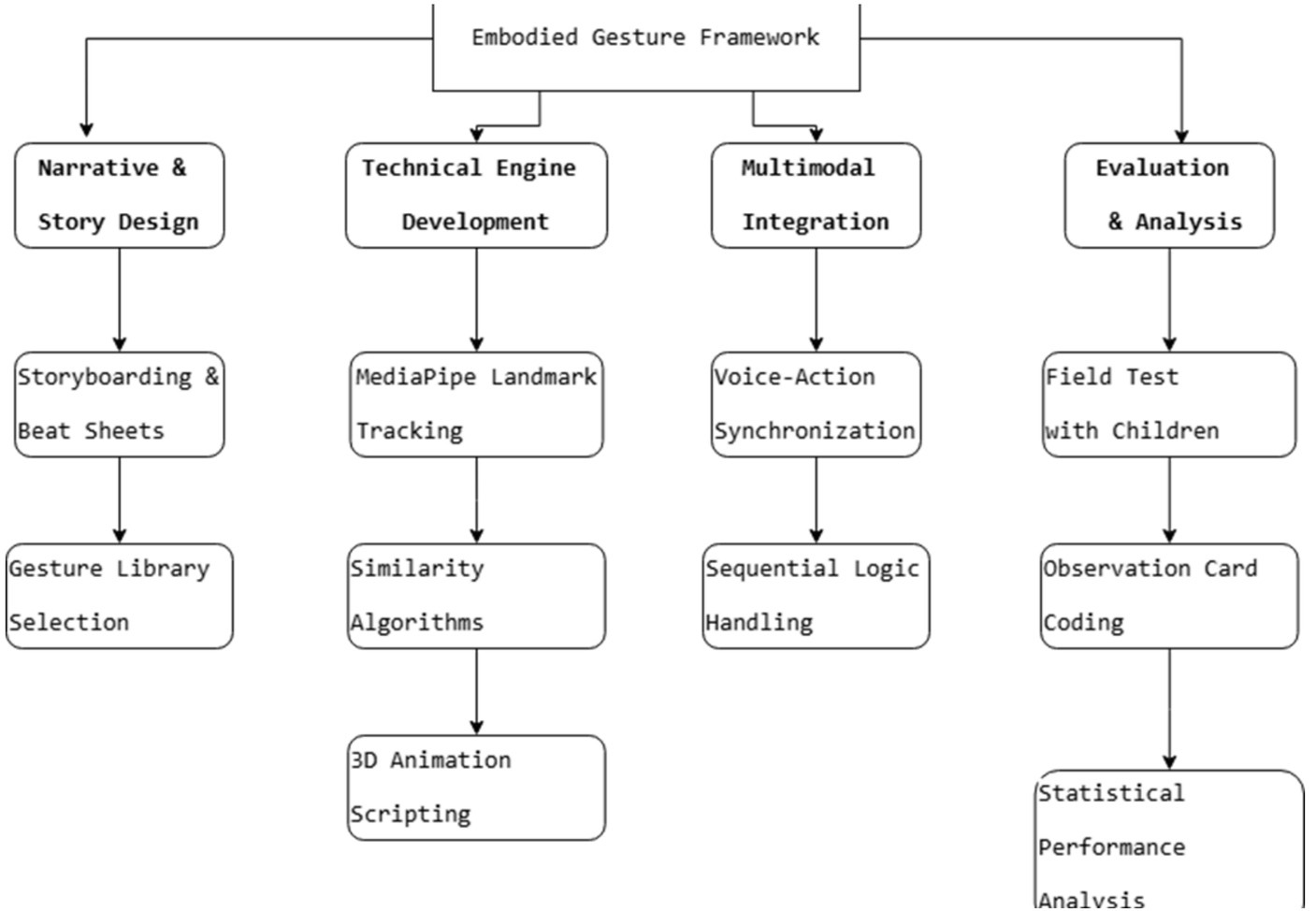
The system operates through the following workflow:

1. Student mimics a STEM concept with their hands
2. Camera captures the movement and MediaPipe extracts 21 hand landmarks
3. Late Fusion model classifies the gesture (85–95% accuracy)
4. System checks the gesture against the correct concept
5. Success unlocks the next story and triggers voice feedback; failure shows a hint
6. Interaction is logged anonymously for analysis (attempts, result, time)



Advance Query-Driven Voice Interaction for STEM Learning in Primary Education

Work Breakdown Structure



AR STEM Story desgin and Embodied Gesture Interaction

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Advance Query-Driven Voice Interaction for STEM Learning in Primary Education

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Advance Query-Driven Voice Interaction for STEM Learning in Primary Education

Problem Definition

Primary school STEM education mainly depends on textbooks and static images, which makes it difficult for students to understand abstract scientific concepts such as:

- Human body structures
- Planetary systems
- Cell structures

most current systems only support touch-based interaction.

Students cannot naturally ask questions while exploring learning content, which reduces engagement and limits deeper understanding.

Advance Query-Driven Voice Interaction for STEM Learning in Primary Education

Research Gap

- Existing AR learning systems mainly focus on visual interaction only.
- Limited support for voice-based question answering during learning.
- Lack of context-aware responses linked to AR educational content.
- Minimal integration of interactive and intelligent learning assistance.
- Need for a system that combines AR visualization with AI-driven voice interaction to enhance STEM learning.

Feature	Voice2Action	AR-Classroom	ECA Storytelling	Kenoui et al.	Dalim et al.	Your System
Voice interaction	✓	✓	✓	✓	✓	✓
Question-driven (not command)	✗	Partial	✓	✗	✗	✓
Intent classification + entity extraction	✓	✗	✗	✗	✗	✓
Coordinated AR visual response	✗	✗	✗	✗	✗	✓
Spoken explanation output	✗	Partial	✓	✗	✗	✓
Primary school children (6–11)	✗	✗	✓	✗	✓	✓
STEM model exploration	✗	✓ (math)	✗	✓ (biology)	✗	✓
Dual synchronized output	✗	✗	✗	✗	✗	✓

Advance Query-Driven Voice Interaction for STEM Learning in Primary Education

Contribution to the Overall Solution

- Enables voice interaction with AR learning content
- Provides instant explanations for STEM concepts
- Improves student engagement and understanding
- Supports an interactive learning experience

Novelty of the Component

- AR + Context-Aware Voice Assistant integration
- Voice questions while exploring 3D models
- AI-based speech recognition
- Multimodal learning (visual + voice interaction)

Advance Query-Driven Voice Interaction for STEM Learning in Primary Education

Research Objective

Develop a query-driven voice interaction system that allows students to ask questions naturally while interacting with AR/MR educational content.

Sub-Objectives

- Implement speech recognition to capture student voice queries.
- Use natural language processing (NLP) to understand questions.
- Generate context-aware answers related to the displayed AR model.
- Improve interactive STEM learning experiences for primary school students.

Advance Query-Driven Voice Interaction for STEM Learning in Primary Education

Research Methodology

This component integrates knowledge from multiple domains:

1. H**uman-Computer Interaction (HCI)**

Designing natural voice-based interaction for children.

2. N**atural Language Processing (NLP)**

Understanding spoken student questions.

3. M**ixed Reality / Augmented Reality**

Providing immersive visualization of STEM concepts.

4. E**ducational Technology**

Enhancing student engagement and learning effectiveness

Advance Query-Driven Voice Interaction for STEM Learning in Primary Education

Technologies Used

The system uses modern technologies including:

- **Speech Recognition APIs** (Google Speech-to-Text / Whisper)
- **Natural Language Processing models**
- **Unity AR / ARCore / ARKit** for Mixed Reality interaction
- **Knowledge-based response systems or AI models**

Advance Query-Driven Voice Interaction for STEM Learning in Primary Education

Functional Requirements

- Students can scan QR codes to view AR STEM models
- System displays interactive 3D learning content
- Students can ask questions using voice commands
- System provides context-aware explanations
- Students can complete quizzes and learning activities
- Teachers can monitor student progress and performance

Non-Functional Requirements

- Usability: Simple and easy interface for young students
- Performance: Fast response for AR visualization and voice queries
- Reliability: System should operate with minimal errors
- Security: Protect student data and user information
- Compatibility: Support for Android mobile devices

Advance Query-Driven Voice Interaction for STEM Learning in Primary Education

Validation and Evaluation

The proposed system will be evaluated through:

- User testing with primary school students
- Measuring learning engagement and comprehension
- Accuracy testing of voice recognition and query responses
- Collecting feedback from teachers and students

Data Availability, Security and Ethics

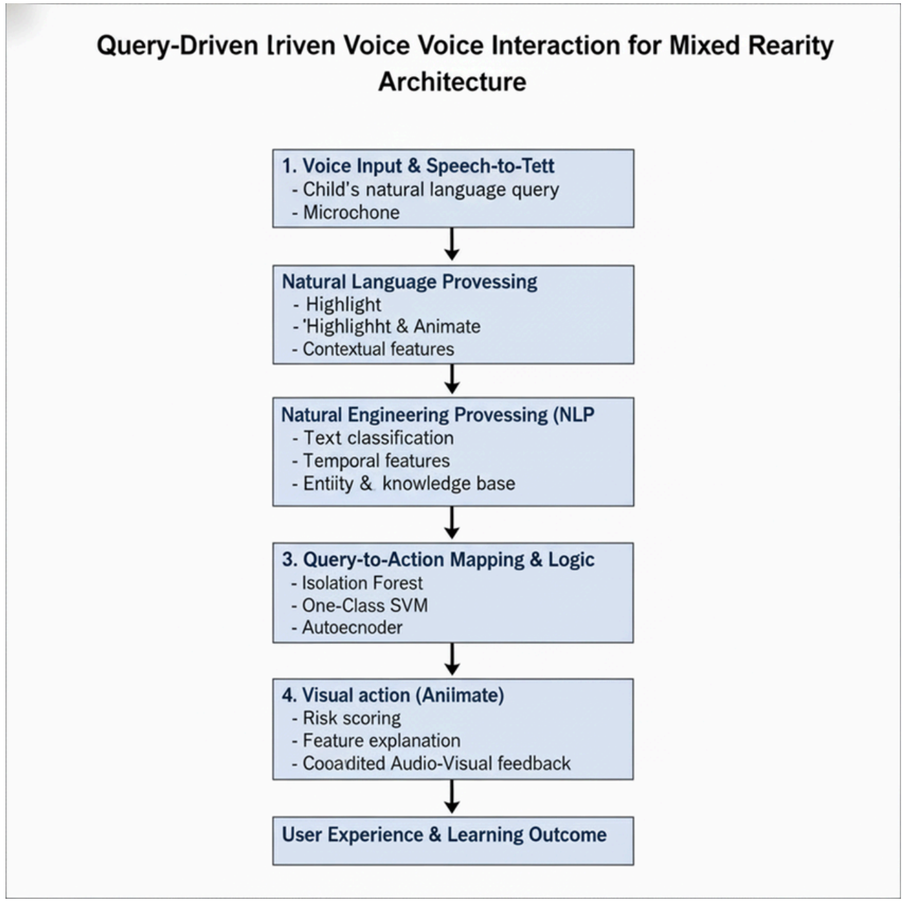
- Student voice inputs will be processed securely.
- No personal student data will be stored without permission.
- Ethical guidelines will be followed during testing with children.

Advance Query-Driven Voice Interaction for STEM Learning in Primary Education

System Overview

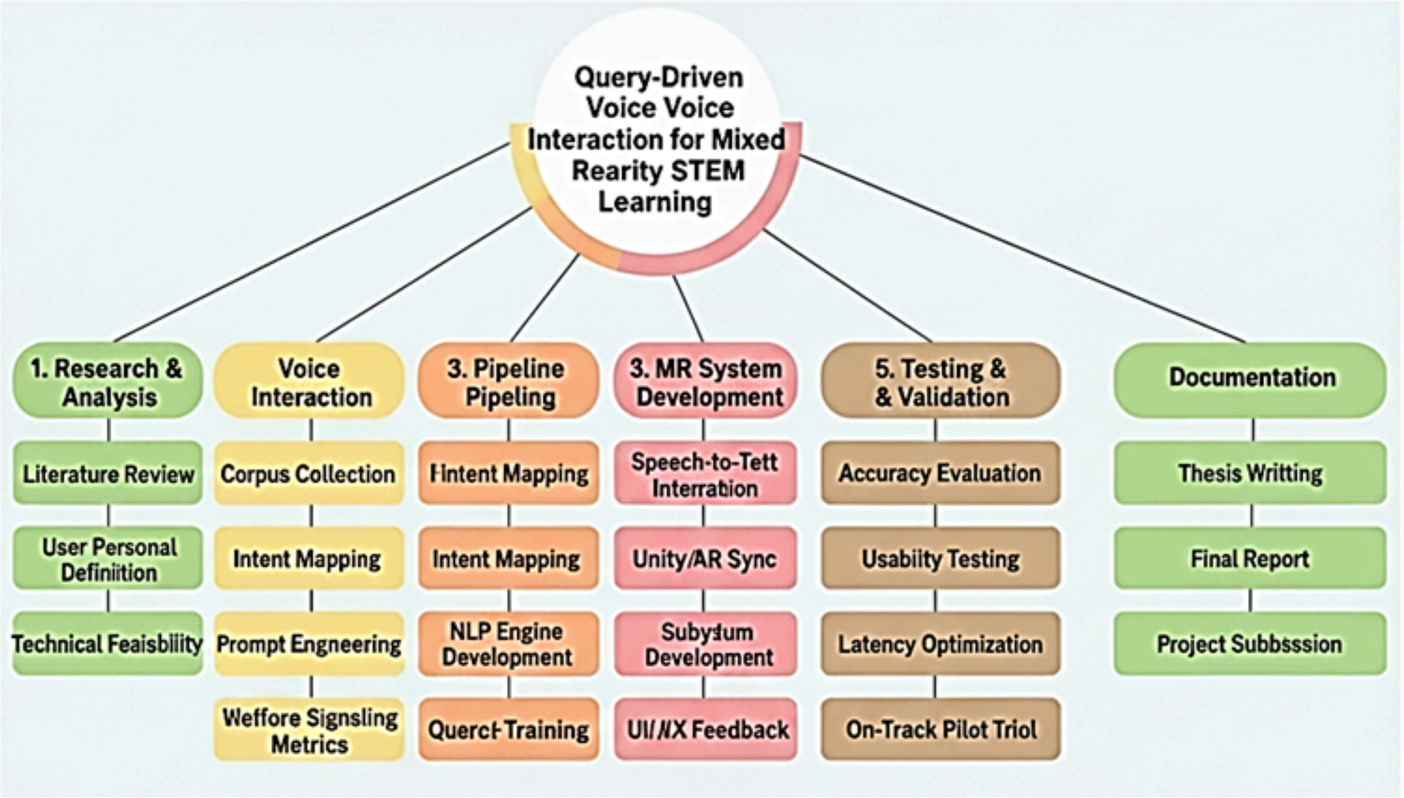
The system operates through the following workflow:

1. Student scans a QR code from a textbook.
2. The AR model appears in the mobile device.
3. The student asks a question using voice.
4. The system converts speech to text.
5. NLP models analyze the query and Highlight the relevant object.
6. The system retrieves a relevant educational explanation.
7. The answer is delivered through voice response or text display.



Advance Query-Driven Voice Interaction for STEM Learning in Primary Education

Work Breakdown



Advance Query-Driven Voice Interaction for STEM Learning in Primary Education

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AI-Enabled Behavioral Quiz Analytics for STEM Learning in Primary Education

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AI-Enabled Behavioral Quiz Analytics for STEM Learning in Primary Education

Problem Definition

In many digital STEM learning systems, students are mainly assessed based on their final quiz scores. While this helps measure performance at a basic level, it does not show how students behave while answering questions or whether they truly understand the lesson.

Key Problems

- Score-only evaluation
- Most systems focus only on correct and incorrect answers.
- No behavior analysis
- Student behaviors such as hesitation, fast guessing, and answer changes are not analyzed.
- Hidden learning difficulties
- Teachers cannot easily identify confidence levels or knowledge gaps.
- Limited feedback
- Students do not receive meaningful feedback based on how they answered the quiz.

AI-Enabled Behavioral Quiz Analytics for STEM Learning in Primary Education

Research Gap

- Existing AR learning systems mainly focus on lesson delivery and final quiz scores.
- Limited attention is given to student behavior during quiz answering.
- Most systems do not analyze response time, hesitation, or answer revisions.
- Current assessment methods cannot clearly identify confidence levels, guessing behavior, or knowledge gaps.
- There is a need for a system that combines AR learning, quiz assessment, and behavioral analytics to provide more meaningful learning insights.

AI-Enabled Behavioral Quiz Analytics for STEM Learning in Primary Education

Contribution to the Overall Solution

- Provides quizzes after the AR lesson to assess student understanding
- Records student quiz behavior such as response time, hesitation, and answer changes
- Helps identify student confidence levels and knowledge gaps
- Supports better feedback before the game activity

Novelty of the Component

- Combines AR learning with quiz behavior analysis
- Analyzes how students answer, not just scores
- Detects confident, hesitant, and guessing behavior
- Generates simple feedback reports

AI-Enabled Behavioral Quiz Analytics for STEM Learning in Primary Education

Research Objective

Develop an AI-enabled behavioral quiz analytics system that provides quizzes after AR lessons, analyzes student quiz interaction behavior, and identifies student confidence levels and learning gaps

.

Sub-Objectives

- Provide short quizzes related to the AR lesson.
- Capture student quiz behavior such as response time, hesitation, and answer changes.
- Analyze quiz behavior to identify confidence, guessing, and learning difficulties.
- Generate a simple feedback report showing student level and knowledge gaps.

AI-Enabled Behavioral Quiz Analytics for STEM Learning in Primary Education

Research Methodology

This component integrates knowledge from multiple domains:

1. **Artificial Intelligence (AI)**
   To support quiz generation and behavior-based analysis
2. L**earning Analytics**
   To study how students answer quiz questions
3. B**ehavior Analysis**
   check response time, hesitation, and answer changes
4. **Educational Technology**
   To improve learning and give better feedback to students

AI-Enabled Behavioral Quiz Analytics for STEM Learning in Primary Education

Technologies Used

The system uses modern technologies including:

- Unity 3D / AR Foundation / ARCore
  for AR-based STEM lessons and quiz interaction in the mobile app
- OpenAI API / Large Language Model
  for generating quiz questions from lesson content
- Unity C# + Behavioral Scoring Model
  for recording quiz behavior and analyzing confidence, hesitation, and guessing
- Python + Firebase / MongoDB
  for processing quiz data, storing interaction logs, and generating feedback reports

AI-Enabled Behavioral Quiz Analytics for STEM Learning in Primary Education

Functional Requirements

- Students can complete quizzes after the AR lesson
- The system records quiz behavior such as response time, hesitation, and answer changes
- The system analyzes student behavior during quiz answering
- The system identifies confidence level, guessing behavior, and knowledge gaps
- The system generates a feedback report after quiz completion
- Teachers can monitor student quiz performance and behavior insights

Non-Functional Requirements

- Usability: Simple and easy interface for primary school students
- Performance: Quick quiz response and feedback generation
- Reliability: Accurate recording of quiz interaction data
- Security: Protect student quiz data and behavioral records
- Compatibility: Support for Android mobile devices

AI-Enabled Behavioral Quiz Analytics for STEM Learning in Primary Education

Validation and Evaluation

The proposed system will be evaluated through:

- User testing with primary school students
- Measuring quiz performance and learning understanding
- Analyzing behavioral indicators such as response time, hesitation, and answer changes
- Checking the accuracy of confidence level and feedback generation
- Collecting feedback from teachers and students

Data Availability, Security and Ethics

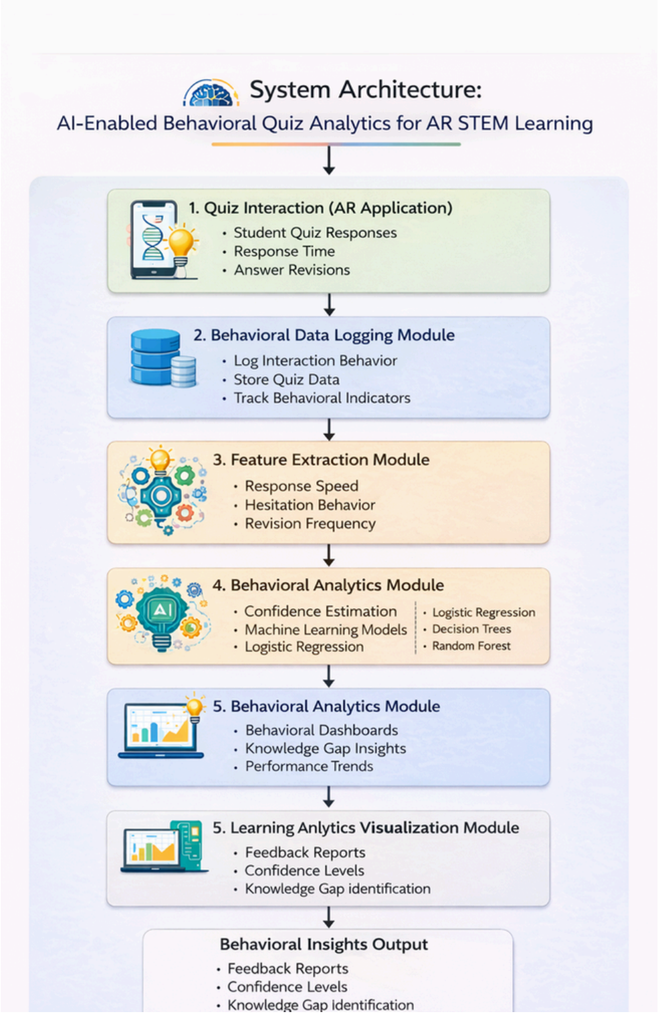
- Student quiz interaction data will be stored securely
- No personal student information will be stored without permission
- All behavioral data will be anonymized before analysis
- Ethical guidelines will be followed during testing with children

AI-Enabled Behavioral Quiz Analytics for STEM Learning in Primary Education

System Overview

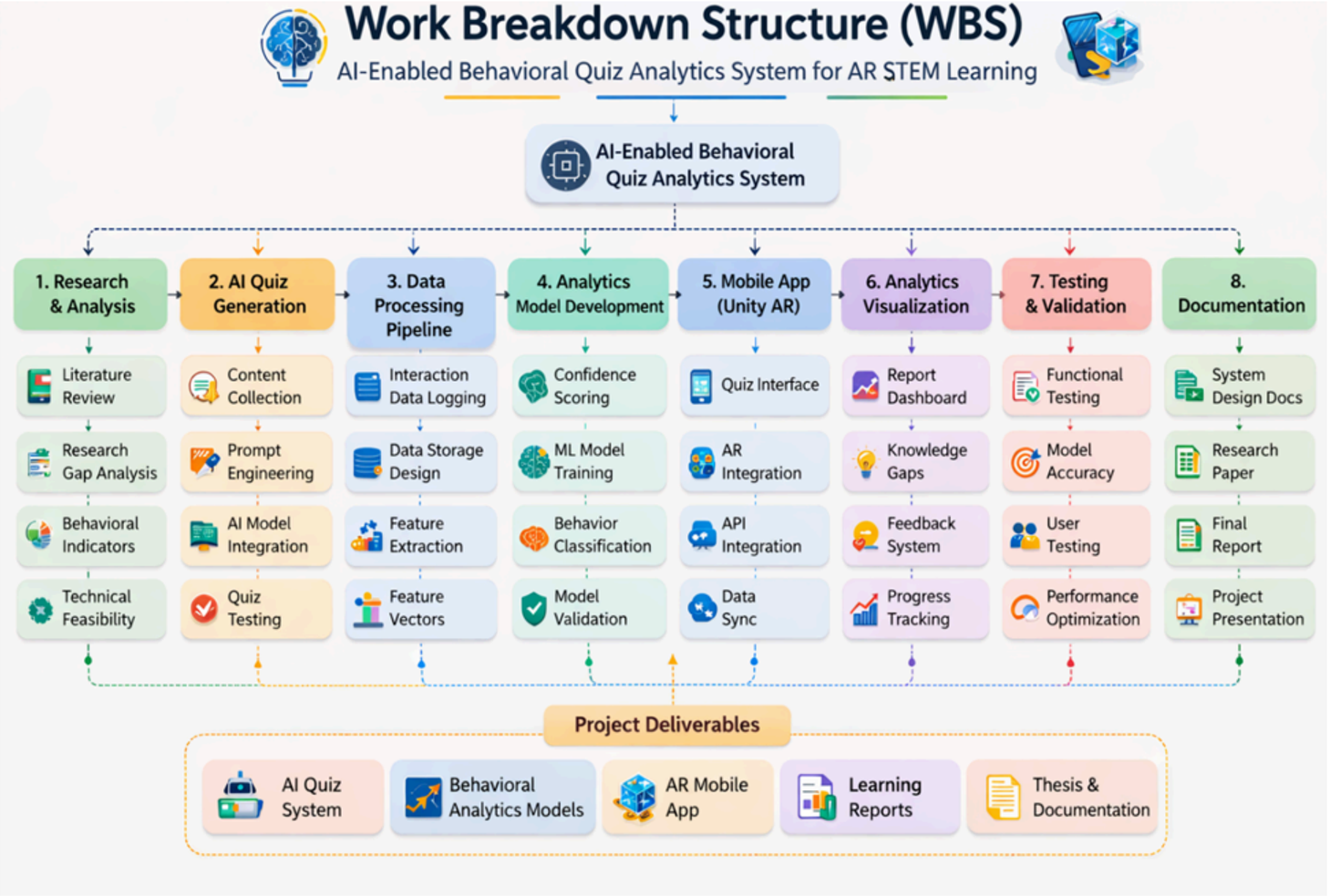
The system operates through the following workflow:

1. The student completes the AR lesson in the mobile application.
2. The system displays a quiz related to the completed lesson.
3. The student answers the quiz questions.
4. The system records quiz interaction data such as response time, hesitation, answer changes, and correctness.
5. The behavioral analysis module processes the recorded quiz data.
6. The system identifies the student's confidence level, behavior pattern, and knowledge gaps.
7. A feedback report is generated and displayed in the mobile application.



AI-Enabled Behavioral Quiz Analytics for STEM Learning in Primary Education

Work Breakdown



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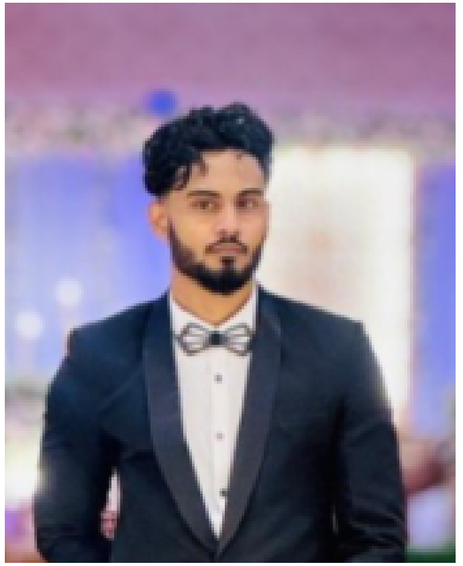

Story-Based Gamified AR Missions for Primary STEM Learning

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Story-Based Gamified AR Missions for STEM Learning in Primary Education

Problem Definition

Primary STEM learning often relies on static teaching methods and passive digital content, which reduces student engagement and active participation.

concepts such as:

- Story-based gamified missions
- Adaptive difficulty based on learner performance
- Meaningful progress trackings

As a result, students may lose interest, participate less actively, and receive limited support for continued learning

Story-Based Gamified AR Missions for STEM Learning in Primary Education

Research Gap

- Many gamified learning systems use only simple points, or quiz-based competition.
- Limited support for story-based mission progression during learning..
- Most current systems do not adapt game difficulty based on learner performance.
- Lack of integration between story flow, AR interaction, rewards, and progress tracking.
- Limited use of adaptive gamified missions to maintain engagement in primary STEM learning.
- Need for a system that combines AR storytelling, gamified missions, rewards, and adaptive progression to improve learner engagement.

Story-Based Gamified AR Missions for STEM Learning in Primary Education

Contribution to the Overall Solution

- Adds a mission-based learning layer to the AR storytelling system
- Encourages active participation instead of passive content viewing
- Supports learner engagement through tasks, rewards, and story progression
- Helps connect story interaction, quiz performance, and learner progress

Novelty of the Component

- Story-based gamified mission integration within an AR learning environment
- Adaptive game recommendation using learner quiz performance
- Reward and progress tracking through points, badges, stars, or achievement status
- Three-level mission difficulty: easy, medium, and hard

Story-Based Gamified AR Missions for STEM Learning in Primary Education

Research Objective

Develop a story-based gamified AR mission module that improves learner engagement by providing interactive tasks, rewards, and progress-based story continuation for primary school students.

Sub-Objectives

- IDesign and develop a story-based gamified mission for one selected primary STEM lesson.
- Implement a basic AR task or activity that allows learners to complete the mission
- Develop a reward mechanism using points, badges, stars, or achievements
- Create a progress tracking feature to display mission completion and learner achievement.

Story-Based Gamified AR Missions for STEM Learning in Primary Education

Research Methodology

This component integrates knowledge from multiple domains:

1. **Requirement identification for story-based gamified missions**
2. **Prototype design for one selected lesson**
3. **Development of rewards, progress tracking, and mission logic**
4. **Adaptive model using quiz score**
5. **Easy, medium, and hard game recommendation**

Story-Based Gamified AR Missions for STEM Learning in Primary Education

Technologies Used

The proposed module uses the following technologies:

- **Unity 3D for developing the gamified AR learning environment**
- **C# for implementing mission logic, rewards, and progress tracking**
- **Adaptive Model to classify learners into easy, medium, and hard levels**
- **Points, Badges, and Stars Mechanism for learner rewards and achievement tracking**

Story-Based Gamified AR Missions for STEM Learning in Primary Education

Functional Requirements

- Students can access a story-based gamified mission within the AR learning environment
- The system displays interactive AR mission content related to the lesson
- Students can complete story-based tasks such as selecting, matching, dragging, or placing objects
- The system provides easy, medium, or hard game levels based on quiz performance
- The system tracks mission status, learner score, and achievement progress

Non-Functional Requirements

- Usability: The interface should be simple, clear, and child-friendly
- Performance: Missions and reward updates should respond quickly
- Reliability: The system should run with minimal errors during gameplay
- Security: Learner score and progress data should be protected
- Compatibility: The module should work within the Unity-based AR system

Story-Based Gamified AR Missions for STEM Learning in Primary Education

Validation and Evaluation

The proposed system will be evaluated through:

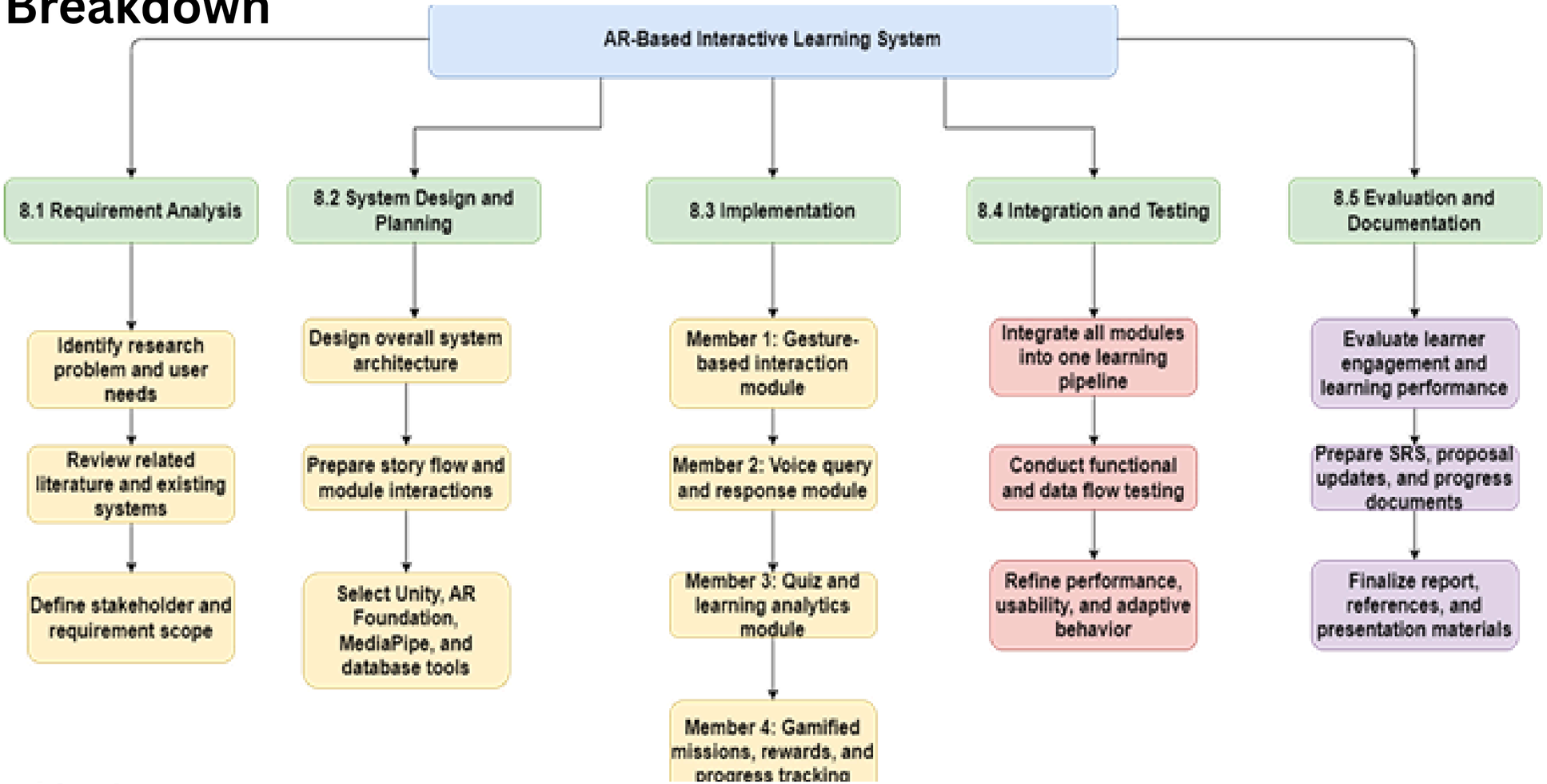
- Functional testing of mission flow, reward logic, and progress tracking
- User testing to observe how learners interact with the story-based mission
- Measuring mission completion, score, and number of attempts
- Evaluating whether the easy, medium, and hard game recommendation works correctly

Data Availability, Security and Ethics

- Learner score, mission status, and progress data will be stored securely
- Personal student information will not be stored without permission
- Ethical guidelines will be followed when testing with children
- Parent, guardian, or school permission will be considered before learner testing

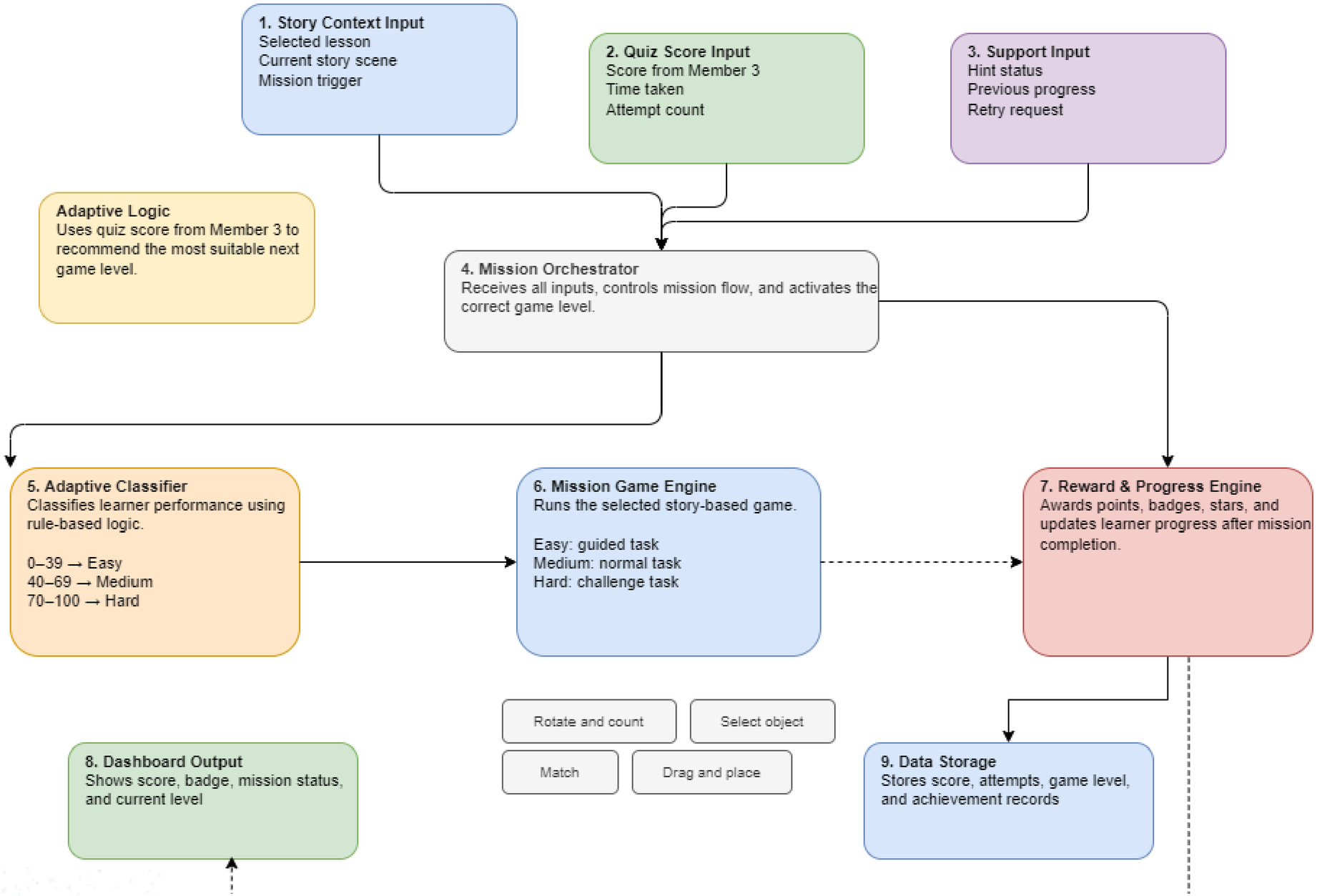
Story-Based Gamified AR Missions for STEM Learning in Primary Education

Work Breakdown



System Architecture — Story-Based Gamified AR Mission Module

Clear architecture view for the individual component



Story-Based Gamified AR Missions for STEM Learning in Primary Education

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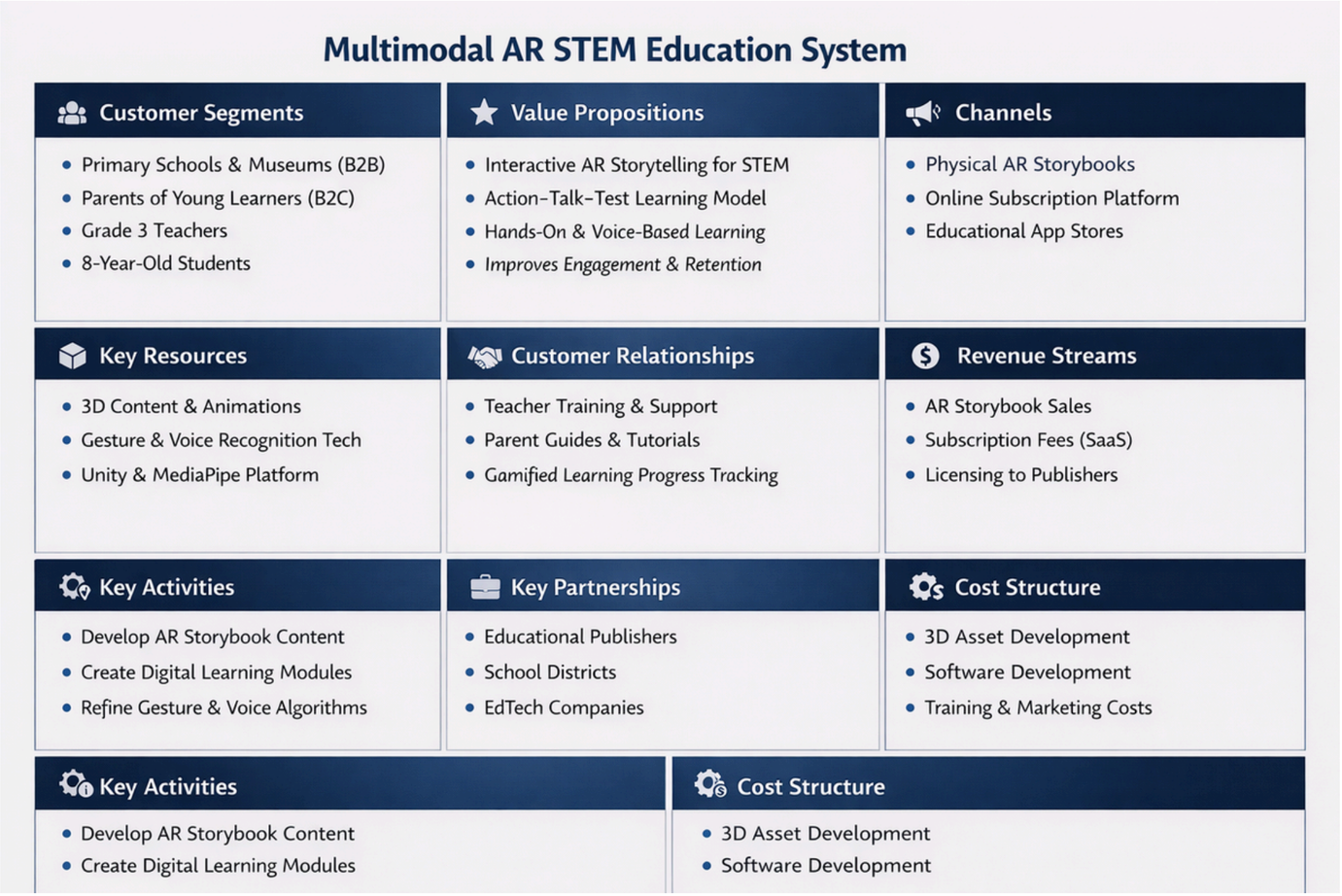
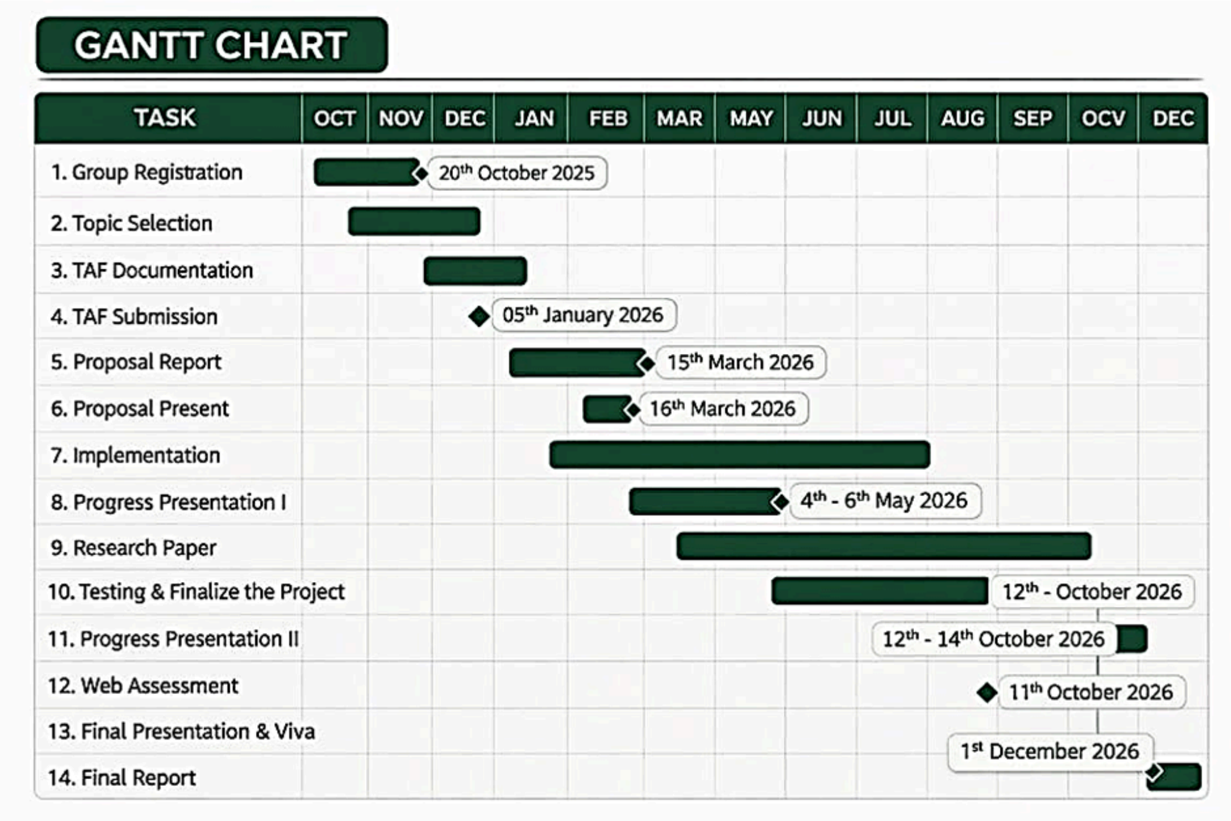


An Adaptive Multimodal AR Storytelling Framework for Primary STEM Education through Embodied Gesture and Voice Interaction



Thank you . . .